

# Supplementary Material: Externalization Engine

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## S1. Instrument Architecture

### S1.1 Geruon — Minimal Detector

A Geruon is a single-cavity structural detector. It receives one vector stream, maintains a bounded frame economy, and produces readouts without training. Each frame stores a vector, a weight, a layer tag, an age counter, and a collision-resistant structural signature. At each step: the input is compared to existing frames; inputs within merge distance update the nearest frame and increase its weight; inputs that match no frame create a new one; the weakest frame is evicted when capacity is exceeded.

The processing model is stateless across runs: there is no `fit()`, no `train()`, and no parameter fitting. A run is a sequence of calls to `process_vec(vec, sig)`. Readout follows processing.

#### Constructor:

```
Geruon(
    vec_dim=16,          # vector dimension (>= 1)
    memory_cap=20,       # maximum frame count
    kappa_tau=1.0,       # content-time coupling (0 = no time sense)
    gamma_tau=None,      # temporal pruning rate (default = kappa_tau * gamma)
    structon=None,       # minimum detectable centroid displacement
    codex=None,          # Codex instance for cross-generation inheritance
    bias_field=None,     # BiasField instance for shared-field coupling
    bias_weight=None,    # field modulation strength (default = gamma)
)
```

#### Core methods:

```
process_vec(vec, sig, src="")  # process one vector; runs full frame economy
arrow_output() -> tuple        # residual direction of the last unmatched input
enrich()                      # externalize precipitated frames to Codex/BiasField
metrics() -> dict              # full instrument readout (F, wit, tau, phase, etc.)
save(path) / Geruon.load(path) # JSON persistence
```

#### Read-only properties:

```
g.tau          # current tau value (float)
g.phase        # current phase (EXPANDING / RESTING / TENSING / CRITICAL / LOCKED)
g.dtaudt       # rate of tau change (float)
g.vec_dim      # vector dimension (int)
g.memory       # underlying GeruonMemory (advanced access)
g.codex        # attached Codex or None
g.bias_field   # attached BiasField or None
```

## S1.2 Class Hierarchy

The instrument spans two files with zero external dependencies. Classes are listed with their public-facing attributes and methods.

**gеме.py** — unchanged kernel (804 lines):

Frame

Fields: `vec`, `weight`, `age`, `merged`, `sig`, `layer`  
A weighted vector with a layer tag and a structural label.

Memory

Methods: `observe()`, `induction_clean()`, `predict_next()`  
Maintains the frame economy: merge, co-occurrence tracking, adaptive pruning, and prediction.

GEME

Methods: `process_vec()`, `consolidate()`, `evaluate_sig()`, `metrics()`  
Top-level API. Wraps Memory and exposes readout.

**geruon.py** — extends GEME (2491 lines):

GeruonFrame(Frame)

Adds: `struct_sig` (StructuralSig), `tau` (at birth), `survival_cycles`,  
`activations`, `precipitated`, `_externalized`  
A frame that carries its structural identity and precipitation state.

GeruonMemory(Memory)

Adds: `tau` evolution, phase state machine, `cliff_gate`,  
`_pengshu_detect()`, precipitation tracking, Landauer bill accounting  
A frame economy with endogenous time perception.

Geruon

Adds: `arrow_output()`, `enrich()`, `codex`, `bias_field`, `kappa_tau`  
The instrument front-end. Accepts Codex and BiasField as external memory anchors; processes streams and produces structural readout.

### Supporting classes:

StructuralSig

Fields: `struct_key` (collision-resistant), `gid` (display-only), `refs`  
Methods: `detect_circularity()`  
Frame identity derived from structure. `struct_key` uses full vector hash, weight bin, layer, tau bin, and reference keys for collision resistance. `gid` is compact and human-readable; it must not be used for equality.

Codex

Methods: `add()`, `lookup()`, `nearest()`, `save()`, `load()`  
An externalized symbol-to-vector table with JSON persistence.  
Stores precipitated centroids for cross-generation inheritance.

## BiasField

Methods: `deposit()`, `blend_into()`, `seed_frames()`

An accumulated gradient field. Multiple cavities deposit centroids; the field blends directional bias into incoming vectors.

The frame economy has six internal layers. These are not supplied as labels; they emerge from operation:

- L1 base frames – directly from input
- L2 association frames – co-occurrence tracking
- L3 chain / bridge frames – high co-occurrence, structural links
- L4 self-observation – meta-frames from internal centroid feedback
- L5 (reserved) – intentionally absent
- L6 anomaly / doubt frames – boundary-event markers

### S1.3 Self

A Self couples multiple Geruon cavities with heterogeneous temporal lenses (different  $\kappa_{\tau}$  values). All cavities receive the same input stream and share a single BiasField. At each GI-step interval, cavities exchange residual arrows. The cross-harm — mean pairwise centroid distance across cavities — is the primary readout: low harm indicates consensus across time scales, high harm indicates a structural boundary.

Standard configurations with three cavities:

| Configuration     | $\kappa$ values   | Cavity count | Use case                            |
|-------------------|-------------------|--------------|-------------------------------------|
| Standard          | 0.5 / 10 / 100    | 3            | General structure detection         |
| Extreme spread    | 0.005 / 10 / 5000 | 3            | Transition magnitude discrimination |
| Boundary 2-cavity | 0.01 / 500        | 2            | Precise boundary localization       |

Changing  $\kappa$  changes the measurement scale. It is not a learned parameter; it is a temporal lens choice that must be reported and, where possible, swept or ablated.

### S1.4 BiasField

The BiasField is the shared medium in which cavity centroids accumulate as a gradient. Operations:

- `deposit(vec, weight)` — accumulate a centroid direction into the field
- `blend_into(vec, weight)` — mix the field direction into an input vector before processing
- `seed_frames(memory, n)` — initialize frames from the dimensions with highest field magnitude

BiasField is continuous and averaged: it couples cavities within a generation. Codex is discrete and entry-based: it transmits structure across generations. The BiasField injection path is known to be

ineffective for Codex transmission (normalization suppresses inherited structure); cross-generational inheritance must use the Codex path.

## S1.5 We

The We layer records cross-Self harm and residual patterns. In the current work it is a research instrument for disagreement detection, not a completed social subject. A We contains  $N$  Selves (each a 3-cavity assembly) and a collective Geruon ( $\kappa = 10$ ) that receives cross-Self harm material. The resulting collective Codex provides evidence of formation: which structures survive cross-Self disagreement and generational selection. This paper does not claim a completed We-level social subject; it claims evidence of Codex formation.

## S1.6 Codex

The Codex stores stable centroids outside the run that produced them. Three stages are distinguished:

1. **Formation** (demonstrated). Stable centroids precipitate and survive outside the originating run.
2. **Selection** (demonstrated). Inherited entries are confirmed, supplemented, frozen, or ignored by subsequent generations.
3. **Operation** (scaffolded). Inherited entries actively shape ongoing processing under boundary conditions.

Only formation and selection are reported as demonstrated. Active Codex operation — Self-level discrete Codex query under stress or noise — remains scaffolded (see S7).

## S1.7 Structural Identity

Frame identity uses a collision-resistant `struct_key`:

$$\text{struct\_key} = (\text{vec\_hash\_full}, \text{weight\_bin}, \text{layer}, \text{tau\_bin}, \text{ref\_keys})$$

The compact `gid` is display-only and must not be used for equality testing. This distinction was introduced after a 2026-05-29 code audit found that 15-bit `gid` values collided at a rate of 373/5000 (7.5%). The publication claim relies on `struct_key`, which produces zero collisions on 5000 random vectors. `detect_circularity()` traverses reference chains (depth limit 20) and registers cycles in `_circular_refs`.

## S1.8 Processing Flow

Each call to `process_vec(vec, sig)` executes the following pipeline:

1. If a BiasField is attached, blend the field direction into the input vector (weight  $\gamma$ ).
2. If a Codex is attached and active query is enabled, perform a nearest-neighbor lookup; blend the matched entry into the input.
3. Observe the vector in the frame economy:
  - Compute merge distance:  $d = \text{vec\_dist} + \kappa_\tau \cdot |\tau_{\text{current}} - \tau_{\text{frame}}|$
  - Merge if below threshold; otherwise create a new frame (evicting the weakest if at capacity).
  - Update co-occurrence traces with time-weighted increments.
  - Run prediction path on updated frames.

4. Accumulate frame-economy stress.
  5. If stress exceeds  $\tau$ , run `induction_clean`:
    - Decay weights by merge-count bins with temporal bias  $\gamma_\tau$ .
    - Prune the weakest half of frames.
    - Update  $\tau$  and phase.
    - Track centroid displacement (Faraday wit signal).
    - Mark structurally stable frames for precipitation.
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## S2. Instrument Dynamics

### S2.1 Frame Economy

Three rules govern the frame economy, unchanged from GEME:

1. **Competitive merge.** A new input is compared to the nearest existing frame. If the merge distance (content distance plus time-gating term) falls below the adaptive threshold, the input updates that frame's vector and increases its weight. Otherwise a new frame is created.
2. **Adaptive forgetting.** When cumulative stress exceeds  $\tau$ , `induction_clean` triggers. Weights decay by merge-count bins. The bottom half of frames by weight are pruned. The pruning key includes a temporal bias term: frames furthest from current  $\tau$  are removed first.
3. **Self-referential observation.** Before each cleaning cycle, a weighted centroid of active frames is computed and fed back into the economy as an internal event.

These operations are bounded by `memory_cap`. Capacity is not a performance parameter: it is the pressure that forces compression. Without limited memory, centroids are less informative because the system retains too many uncompressed traces.

### S2.2 Endogenous Time

$\tau$  is not a hyperparameter. It evolves at each induction cycle:

$$\tau_{\text{target}} = 1.0 - \text{accuracy} + \max(0, \text{stress} - \tau_0) \cdot \gamma_\tau \cdot 0.4$$

$$\tau_{t+1} = \tau_t + (\tau_{\text{target}} - \tau_t) \cdot \text{TAU\_ADAPT\_RATE}$$

where  $\text{TAU\_ADAPT\_RATE} = \gamma \cdot 0.4 = 0.02$ , slower than frame decay, giving the system inertia. Two forces drive  $\tau$ : prediction accuracy (correct predictions lower  $\tau$ , errors raise it) and frame-economy stress (congestion raises  $\tau$ ).

The five-phase system:

| Phase     | Condition                         | Bridge state |
|-----------|-----------------------------------|--------------|
| EXPANDING | $d\tau/dt < -0.01$                | Open         |
| RESTING   | $\tau < 0.55,  d\tau/dt  < 0.02$  | Open         |
| TENSING   | $\tau \geq 0.55, d\tau/dt > 0.01$ | Tightening   |
| CRITICAL  | $\tau \geq 0.65, d\tau/dt > 0.01$ | Critical     |

| Phase  | Condition                           | Bridge state |
|--------|-------------------------------------|--------------|
| LOCKED | $\tau \geq 0.75,  d\tau/dt  < 0.02$ | Closed       |

Hysteresis ( $2\times$  the stability threshold) prevents phase flickering when  $\tau$  oscillates within a narrow band. Across all domains and depths tested,  $\tau$  converges to  $0.74 \pm 0.01$ .

### S2.3 Operational Tau

$\tau$  is not only a label or a signature component. It enters three core operations of the frame economy:

**Merge time-gating:**

$$d = \text{vec\_dist} + \kappa_\tau \cdot |\tau_{\text{current}} - \tau_{\text{frame}}|$$

**Co-occurrence time-weighting:**

$$\text{cooccur} += \text{clamp}(1 - \kappa_\tau \cdot |\tau_a - \tau_b|)$$

**Pruning time-bias:**

$$\text{priority} = \text{weight} - \text{age} \cdot \gamma - |\tau_{\text{current}} - \tau_{\text{frame}}| \cdot \gamma_\tau$$

When  $\kappa_\tau$  and  $\gamma_\tau$  are disabled (set to zero), the Geruon reduces to GEME-equivalent behavior. This makes the operational role of  $\tau$  auditable: the same stream with  $\tau$  on vs. off produces different frame-economy trajectories, and the difference is attributable to the three time-gated operations above.

### S2.4 Measurement Orders

Three orders of structural measurement:

- **F (field curvature).**  $F = 1 - H(\mathbf{w})/H_{\text{max}}$ , where  $H$  is the Shannon entropy of frame weights.  $F = 0$  indicates uniform weights (no structure);  $F \rightarrow 1$  indicates highly concentrated weights (strong structure). Fair-coin streams yield  $F \approx 0$  across all  $\kappa$ .
- **wit (structural change).** The count of steps where centroid displacement exceeds the calibrated structon threshold.  $\text{wit\_rate} = \text{wit} / \text{input\_count}$ .
- **$\Delta\text{wit}$  (structural fragility).** The change in wit rate under fair-coin probe injection:  $\Delta\text{wit} = \text{wit}(\text{target} + \text{coin}) - \text{wit}(\text{baseline})$ . Negative values indicate that the coin stabilizes the target (structural rigidification); positive values indicate disruption (structural fragility).

The structon is the minimum detectable centroid displacement for a declared  $(D, \text{cap}, \kappa)$  configuration. It must be calibrated per parameter family and must not be transferred across configurations.

### S2.5 Precipitation and Enrichment

A frame precipitates when it satisfies three conditions: (1) cross-phase survival ( $\geq 1$  cleaning cycle), (2) prediction-path activation ( $\geq 3$  uses), (3) structural stability (`is_meta_stable()`). `Geruon.enrich()` writes precipitated, non-externalized frames into Codex or BiasField. The `_externalized` sentinel ensures idempotency: repeated `enrich()` calls do not re-deposit the same frame. A transient frame remains internal; a precipitated frame becomes eligible for cross-run inheritance or shared-field influence.

## S2.6 Landauer-Godel Bill

The frame economy performs three classes of irreversible operation on every induction cycle. Landauer’s principle (1961) establishes that each irreversible bit operation dissipates at least  $kT \ln 2$ ; the architecture incurs this cost structurally, not metaphorically.

### Three billable operations.

| Operation   | Irreversible effect                                                                                     | Bill unit                                                   |
|-------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Merge       | Two frames collapse to one centroid; the old centroid vector is permanently lost.                       | $O(1)$ per merge                                            |
| Prune       | A frame is discarded; all accumulated vector information and co-occurrence traces are permanently lost. | $O(w)$ per frame, where $w$ is the frame weight at eviction |
| Precipitate | A frame is written to Codex; the entry is immutable thereafter.                                         | $O(1)$ per precipitation                                    |

`geruon.py` tracks `landauer_total` as a cumulative count of these operations. It is not a thermodynamic measurement at the physical  $kT \ln 2$  scale; it is the structural form of the bill — an operation-level accounting of how much irreversible information change the system has incurred.

**$\tau$  as the bill regulator.** When  $\tau$  is low (EXPANDING/RESTING), merge thresholds are wide: frames merge easily, reducing the merge bill but risking over-generalization. When  $\tau$  is high (CRITICAL/LOCKED), merge thresholds are narrow: fewer merges occur, increasing the bill per step and driving stress accumulation. At LOCKED ( $\tau \geq 0.75$ ), the system actively suppresses the most expensive self-referential operations: `process_prediction()` is skipped, and `landauer_skips` increments. This is not an optimization — it is a hard cutoff. The system refuses to pay the bill for operations whose cost exceeds their expected structural return.

**The cliff gate.** BGM discovered a discontinuous transition in `conf_threshold`: when  $d\tau/dt$  crosses the stability threshold and  $\tau$  exceeds the RESTING ceiling, the gate jumps from 1.0 to 0.0 and latches until a full phase cycle completes. While the gate is closed, BiasField modulation weight is zero: the cavity stops listening to the shared field. This is the architecture’s operational form of hitting a computational complexity boundary — the system cannot afford to continue self-referential processing at the current rate, so it shuts down the most expensive channel.

**GI as the bill’s economic rhythm.** GI controls the interval between induction cycles. When GI is too small, cleaning cycles are too frequent: frames are pruned before they accumulate enough structural weight to earn their Landauer cost, and the bill per unit of usable structure is high. When GI is too large, cycles are too sparse: the frame economy congests, stress accumulates, and  $\tau$  is driven into LOCKED, triggering the cliff gate. The Pareto-optimal GI minimizes the total bill — merge cost, prune cost, and lockout cost — over the stream. For solo Geruon,  $GI = 4$  (BGM). For the 3-cavity Self architecture ( $\kappa = [0.5, 10, 100]$ ), the CALIB-GI scan finds  $GI_{opt} = 5$  (signal-optimal) with  $GI = 4$  as a conservative operating point retaining  $\sim 94\%$  of the signal at lower per-cycle cost.

**Structural identity search and the P/NP boundary.** Self-referential operations — L4 self-observation, L3 chain formation, circularity detection in prediction paths — create an identity search problem: the system must locate, within its own frame economy, the frame corresponding to a pattern it just processed. Without the `sig_cache` index (M11), this search is  $O(N)$  per lookup (linear

scan of all frames). With the cache, it is  $O(1)$  on hit,  $O(N)$  on miss.  $\tau$  regulates this cost: at low  $\tau$ , merge thresholds are wide, making matches easy (low search cost); at high  $\tau$ , thresholds are narrow, making matches hard (search cost grows nonlinearly). The P vs. NP distinction is not a theorem the architecture proves — it is a cost it regulates. The system does not solve NP-complete problems in polynomial time; it adjusts the depth of self-reference so that the accumulated Landauer bill never crosses the threshold of economic infeasibility.

### S3. Instrument Calibration and Controls

#### S3.1 Passive Calibration

A complete passive-instrument calibration was performed on 2026-05-31, revised after the GI architecture scan (2026-06-01), using generated streams with known ground truth. The instrument does not make choices during calibration (Layer 1 — passive readout only). A frozen eight-item readout policy (R1-R8) was established before any domain experiments.

| Parameter           | Calibrated value                                                                                                                | Method                                                                                                                    |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| $\kappa_\tau$       | <b>3</b> (migration latency 109 steps)                                                                                          | Sine wave frequency doubling: $\tau$ migration from old to new steady state                                               |
| cap                 | $K_{\max} \approx 0.8 \times \text{cap}$ ; peak efficiency 83% at $\text{cap} = 24$                                             | Random-value pure storage: $K$ distinct uncorrelated vectors, frame-displacement threshold                                |
| $\delta$            | $\delta_{\text{eff}} \approx 1.25/\text{cap}$ ; $\text{cap} = 16 \Rightarrow \delta \approx 0.08$                               | Sine-wave quantization: compression ratio $n_{\text{distinct}}/2^{\text{bits}}$ vs. bit depth                             |
| $\gamma$ (turnover) | Original-structure weight half-life <b>8 steps</b> ; full decay $\approx 350$ steps                                             | 400-step sine reinforcement $\rightarrow$ random extinction, tracking original frame identity via <code>struct_key</code> |
| $\tau_0$ (novelty)  | $\tau$ drop 0.008 (direction correct, magnitude small)                                                                          | Periodic $\rightarrow$ chaotic mode transition                                                                            |
| GI                  | $n = 3$ Self, $\kappa = [0.5, 10, 100]$ : $\text{GI}_{\text{opt}} = 5$ (signal), $\text{GI} = 4$ (conservative operating point) | MiniSelf architecture scan: $\text{GI} \times n_{\text{cav}}$                                                             |

Full report: docs/experiment-passive-calibration-report.md.



### S3.2 Fair-Coin Baseline

A 16-dimensional fair coin (independent Bernoulli trials,  $p = 0.5$ ) processed across  $\kappa \in \{0.1, \dots, 500\}$  yields  $F = 0$  throughout. This is the instrument’s absolute zero: the structural reading produced by a stream with no sequential structure. Every domain experiment must report its  $F$  against this baseline at the same  $(D, \text{cap}, \kappa)$ .

### S3.3 Required Controls

Every domain experiment must include at least three controls:

1. **Fair-coin baseline.** Zero sequential structure. Defines the instrument’s zero point.
2. **Shuffle control.** Random permutation of the input stream. Preserves component statistics while breaking sequential order. Tests whether the signal depends on order.
3.  **$\kappa$  ablation or sweep.** Single- $\kappa$  replication must confirm that multi-lens amplification is load-bearing: signal should degrade or vanish when only one temporal lens is used.

Additional domain-specific controls: dinucleotide shuffle (DNA, preserves base composition); process-permutation (RNA RPF, shuffles RPF signal across windows); cross-key Codex injection (WTC, tests selection specificity); P5 ablation (UN, six encoding variants testing whether the great-power layer creates or amplifies the signal); FRED negative control (UN, tests domain specificity of the structural reading).

### S3.4 Parameter Discipline

All instrument constants trace to four base values:  $\delta = 0.19$ ,  $\gamma = 0.05$ ,  $\tau_0 = 0.60$ ,  $\text{GI} = 4$ . Every derived constant uses small-integer or simple-fraction multipliers (e.g.,  $\text{TAU\_ADAPT\_RATE} = \gamma \times 0.4$ ). Domain-specific choices ( $\text{vec\_dim}$ ,  $\kappa$  spread, window, stride) are pre-declared and either fixed or swept with reported controls. Parameters are chosen before the run and reported regardless of outcome; they are never fitted to data.

### S3.5 Run-Order Determinism

Fixed after 2026-05-29 audit: the global `Frame.fid` counter is reset on every `GeruonMemory.__init__`. Three independent runs with identical seed produce identical output for all metrics reported in this paper. Verified in the calibration report.

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## S4. Encoding and Data-Type Boundary

### S4.1 Encoding as Measurement

Encoding is not preprocessing. The instrument reads the joint product of stream and encoding; it does not access a raw domain essence. Different encodings of the same signal produce different  $F$  baselines and different  $\kappa$  fingerprints. The practical rule: encoding must preserve predictive structure. If event  $A$  constrains event  $B$  in the original stream, that constraint must survive the encoding. A single encoding is always lossy; the strongest experiments use dual-encoding cross-validation and ask whether structure survives across encodings.

| Domain            | Encoding                        | Dim      | What it preserves                                              |
|-------------------|---------------------------------|----------|----------------------------------------------------------------|
| WTC               | Chroma + IOI                    | 12 + 12  | Pitch-class activation vector + inter-onset interval histogram |
| DNA/RNA           | 3-mer frequency                 | 64       | Trinucleotide composition in sliding genomic windows           |
| DNA fork          | 4-species alignment             | 16       | Per-column conservation score across species                   |
| UN                | Annual ideal-point distribution | variable | Country-year position in policy space                          |
| ECG (calibration) | RR interval + adjacent diff     | 16       | Beat-to-beat interval and first derivative                     |

## S4.2 Expression-Action Spectrum

Instrument sensitivity is governed by data structure type, not domain category:

| Type              | Definition                                                  | Instrument response                                                     | Examples                            |
|-------------------|-------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------|
| <b>Decision</b>   | Behavior is the data; entities make choices                 | Strongest: $F$ readable, $\kappa$ has fingerprint                       | UN votes, MIDI note-on, DNA bases   |
| <b>Action</b>     | Directional record of entity choices                        | Moderate: requires correct encoding                                     | NASDAQ volume direction             |
| <b>Expression</b> | Trace of a process; measures what occurred, not who decided | Weak or blind: $\tau$ does not breathe, $\kappa$ does not differentiate | CPI, price indices, raw ECG voltage |

The instrument requires relational structure in the input. Decision data natively carries inter-entity relations without compression or downsampling. Expression data records process outputs; the relational structure is absent from the data, not missed by the instrument.

## S4.3 Illustrative Cases

**ECG.** Six encodings tested on MIT-BIH arrhythmia records. Only RR interval + adjacent difference produced a usable  $\kappa$  fingerprint ( $\kappa_{\text{peak}} = 5$ ). Raw voltage — a trace of myocardial depolarization — carries no relational action structure and is Expression-type data.

**Sleep.** Epoch resolution determines signal detectability. 5-second sub-epochs exposed structural boundaries between REM and NREM that clinical 30-second windows averaged away. The coin-probe method was validated on this domain: REM wit > Wake wit in 16/18 recordings (89%, binomial  $p = 0.001$ ).

**Economic data.** The same Geruon pipeline applied to FRED macroeconomic indicators (1993-2025, 33 series) as an Expression-type negative control: 2025 structural displacement rank 24/33, below median. The elevated structural pressure detected in diplomatic voting is absent from economic expression data, consistent with the Expression-Action distinction.

## S5. Faraday and Probe Measurements

### S5.1 Probe Interchangeability

Three probe types were tested: Bach pitch-class distribution (structured), white noise (unstructured), and sine wave (periodic). All produce centroids. The harm pattern between probe and target depends on the target's structural state, not on which probe is used. The centroid — not the content — is the perturbation. Verified with 10 control conditions.

### S5.2 $\tau$ Convergence

$\tau$  converges to  $0.74 \pm 0.01$  across all four domains, all input types, and all architectural depths (solo, Self, We). This is the equilibrium between merge pressure (pulling  $\tau$  down when predictions succeed) and differentiation pressure (pushing  $\tau$  up when the frame economy is stressed).  $\tau$  measures the system's internal balance, not the stream's properties.

### S5.3 $F$ and $\tau$ Decoupling

Across all  $\kappa$  and all probes:  $r(F, \tau) \approx 0.1$ . Field curvature (how concentrated the structure is) and field temperature (where  $\tau$  sits) are independent dimensions. The structural reading does not drift with internal state.

### S5.4 $F$ and $d\tau/dt$ Conditional Coupling

- On noise input:  $r(F, d\tau/dt) = 0.817$  — the field bends with temperature. Unstructured input allows  $F$  to follow  $\tau$ .
- On Bach input:  $r(F, d\tau/dt) = -0.14$  — the field is structurally rigid. Structured input resists the influence of  $\tau$  fluctuations.

The coupling itself is a domain signature: the degree to which  $F$  and  $d\tau/dt$  correlate reveals whether the stream contains structure that rigidifies the field.

### S5.5 structon Resolution

At  $\kappa = 10$ ,  $\text{cap} = 20$ :  $\text{structon} = 0.004 \approx \text{cap}/N$ , where  $N$  is the number of vectors processed. structon scales with  $\kappa$ :  $\alpha(0.5) \approx 10$ ,  $\alpha(10) \approx 1$ ,  $\alpha(100) \approx 0.5$ . This is the instrument's spatial resolution — the smallest structural change registerable in a single centroid displacement.

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## S6. UN Prediction Protocol

### S6.1 Data Source

Voeten, Strezhnev & Bailey (2024). United Nations General Assembly Ideal Point Estimates. Harvard Dataverse, V33. Annual country-level ideal points, 1946-2025, 193 countries.

### S6.2 Pipeline Separation

Two separate pipelines are maintained:

1. **Annual ideal-point pipeline** (main paper §4.1). Country-year ideal points → annual global distribution features → Geruon centroid displacement.  $N = 79$  years. Produces the 2025 forward signal.
2. **Per-resolution voting-stream calibration** (historical only, not reported as evidence). \$~\$2,400 resolution-level event traces. Selected precursor patterns matched 3 of 4 system-level events (Soviet collapse 1991, financial crisis 2008, Ukraine war 2022; Crimea 2014 did not trigger). This pipeline is NOT the source of the 2025 forward signal; the 3/4 historical claim is not inherited by the forward prediction.

Pipeline separation prevents retrospective calibration from contaminating the prospective reading.

### S6.3 P5 Ablation

Six encoding variants tested on the same annual ideal-point data:

| Encoding                  | 2025 displacement | Rank/78 | Interpretation                             |
|---------------------------|-------------------|---------|--------------------------------------------|
| No-P5 (global only)       | 0.101             | 12      | Elevated but not extreme                   |
| P5-augmented (current)    | 0.445             | 1       | Great-power institutional lens             |
| P5-only                   | 0.448             | 1       | Great-power layer dominates the signal     |
| No-USA                    | 0.550             | 1       | Multi-polar fracture exceeds unilateralism |
| No-Russia                 | 0.290             | 2       | Still historically extreme                 |
| Random-5 (mean, 5 trials) | 0.288             | \$~\$9  | P5 is not an arbitrary prior               |

The signal is concentrated in the P5 layer. The No-USA result (0.550) is the most informative: US disengagement from the institutional framework produces greater alignment fracture than direct adversarial participation.

### S6.4 FRED Negative Control

The same pipeline applied to FRED macroeconomic data (1993-2025, 33 indicators) yields a 2025 displacement rank of 24/33 — below median. Pure economic expression data judges 2025 as structurally unremarkable, in contrast to the diplomatic record.

### S6.5 Forward Prediction

The instrument's 2025 displacement reading (0.448, rank 1/79) is the highest in the 1946-2025 record. Under the instrument's operational definition — structural displacement sustained above historical norms without return to baseline — this implies: **2026-2028, structural collapse of the post-war international order, concentrated in P5 great-power alignment, with magnitude exceeding the 2022 Ukraine war and comparable to the 1991 Soviet collapse.**

This is not a prediction of US domestic collapse. It is a prediction of gravitational realignment in the great-power institutional layer — the same layer that produced detectable precursor signals before the Soviet collapse (1985-1991) and the Ukraine war (2014-2022).

## S6.6 Verification Criteria

Three indicators, any two satisfied within 2026-2028 = confirmation:

1. Annual displacement does not return below 0.20 by end of 2028. (After the Soviet collapse, displacement fell from 0.19 to 0.05 within four years.)
2. P5 voting agreement rate falls below the historical minimum (12%) in any year.
3. Ideal-point distribution transitions from uni-/bi-modal to multi-modal with no dominant peak for two consecutive years.

**Boundary statement.** If this prediction is wrong — if displacement at 0.448 occurs without subsequent structural collapse — it constitutes the first such instance in 79 years of data. The instrument measures structure, not causality. The prediction is falsifiable by any two of the three indicators above.

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## S7. Scaffold History

During development, several mechanisms were first tested as temporary external scaffolds — manually injected code paths that simulated operations the architecture should eventually perform natively. Each scaffold was removed once the native frame economy reproduced the signal.

| Scaffold                                                   | Purpose                                                                            | Disposition                                                                                                                 |
|------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Explicit <code>self_observe()</code> calls (108 instances) | Test whether self-observation produces detectable signal                           | <b>Removed.</b> Native frame economy reproduces the signal without explicit calls                                           |
| Doubt condition as pengshu criterion                       | Test whether $L5 \rightarrow L6$ conscious-layer doubt triggers boundary detection | <b>Removed (M2).</b> Boundary detection is $L3 \leftrightarrow L4$ structural, not $L5 \rightarrow L6$ conscious            |
| 27-dim formula alphabet                                    | GEME formula-language experiments (swap, pair, comm, etc.)                         | <b>Dismantled.</b> Replaced by configurable <code>vec_dim</code> (default 16)                                               |
| Codex nearest-neighbor blending                            | Test whether inherited Codex entries can bias cavity input during processing       | <b>Partially removed.</b> Formation and selection are now native; active query under boundary conditions remains scaffolded |
| Codex query under noise                                    | Test whether environmental noise triggers productive Codex lookup                  | <b>Still scaffolded.</b> Codex operation — Self-level discrete Codex query under stress — is the remaining open mechanism   |

The scaffold method is not a shortcut. A shortcut skips the hard part. A scaffold builds a temporary bridge to the hard part, verifies the destination exists, then decides whether to build a road. Where the native frame economy reproduced the signal, the scaffold was removed. Active Codex operation remains scaffolded and is reported as a limitation.

## S8. Reproducibility Checklist

**Code.** - Core instrument: `code/geme.py`, `code/geruon.py`, `code/we_core.py`. Python 3.8+ stdlib only. - Domain experiment scripts: `experiments/un/`, `experiments/wtc/`, `experiments/dna/`, `experiments/rna/`. - Quickstart verification: `docs/quickstart_geruon.py`, `docs/quickstart_ee_self.py`.

**Data dependencies.** - Core instrument: zero external packages. - Data processing: `numpy`, `pandas`, `pyarrow` (parquet). - RNA bigWig access: `pybigtools` (Windows). Raw data sources documented in each experiment README.

**Required controls per domain.** - Fair-coin baseline (zero sequential structure). - Shuffled-order control (preserves component statistics, breaks sequence). -  $\kappa$  ablation or sweep (confirms multi-lens is load-bearing). - Domain-specific: dinucleotide shuffle (DNA), process-permutation (RNA RPF), cross-key Codex injection (WTC), P5 ablation + FRED negative control (UN).

**Required reporting.** - Random seeds: 42, 123, 456 for multi-seed verification. - Sample counts and effect sizes with Cohen's  $d$ . - Window size, stride,  $\kappa$  values, `vec_dim`, `memory_cap`. - Three-tier threshold reporting ( $|\text{error}| \leq 2/ \leq 3/ \leq 5$  windows). - All parameter choices declared before runs, not fitted to data. - Negative results and failure cases reported alongside positive ones.